

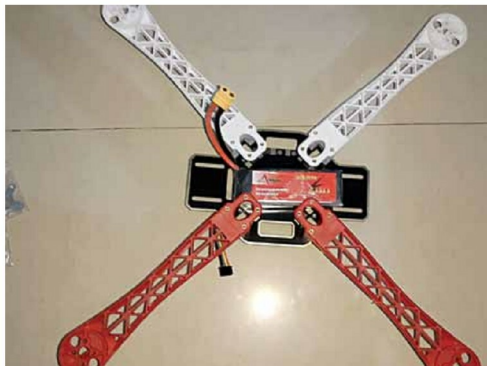


Fig. 3: Four arms of quadcopter attached to the central frame



Fig. 4: The KK 2.2 flight controller mounted on the main frame



Fig. 5: Quadcopter drone with ESC module mounted on each arm

propeller's rotation should provide sufficient thrust to lift the drone. To achieve this, the motors need to be setup in the following manner:

- Front left motor should rotate clockwise (CW)
- Front right motor should rotate counter-clockwise (CCW)
- Rear left motor should rotate counter-clockwise (CCW)
- Rear right motor should rotate clockwise (CW)

So, we need to install the motors and propellers accordingly as shown in Fig. 2.

Electronic speed controller

Each motor in quadcopter is attached to a circuit called electronic speed controller (ESC). The quadcopter's flight controller sends information to the motors via ESC. Each ESC module must be rated at least 10A more than the motor. Suppose your motor is going to consume 20 amperes on full throttle, then the ESC should be rated 30A or more.

The Kv rating of motors refers to the constant velocity of a motor (it is not kV for kilo-volt). Low Kv rating means a lower rpm of the motor. This means a longer propeller would be required to get sufficient thrust to lift the drone. With higher Kv rating of the motors, smaller propellers can be used.

In this project, 1000Kv brushless motors are used along with 30A electronic speed controllers. Since Kv rating is low, we need at least 25.4cm propeller to lift the F450 drone body easily.

In case a smaller drone frame like QAV250 is used with smaller 12.7cm propellers, the same motors will not be able to lift of the drone. To get adequate amount of thrust from these propellers, we will need higher Kv rating, such as 2000Kv, motors.

Radio transmitter and receiver

FlySky FS-CT6B transmitter and receiver set is used in this project. These are pre-programmed and pre-bound. If you wish to change the radio transmitter or receiver then you will need to bind them first.

Radio transmitter requires some settings/configurations to control the drone. First, you need to bind your transmitter to your receiver, which enables your transmitter to control the